Ahmedabad University

PSY 310 - Lab In Psychology

Motor Sequence Learning Task

Section - 1

Submitted to faculty: Prof. Nithin George

Date of Submission: 07.11.26

Student Details

Roll No.	Name of the Student	Name of the Program
AU2420024	Hetansh Jalan	B.A Psychology

2026-2027 (Monsoon Semester)

GitHub Link - https://github.com/HetanshJalan/Psychology-Experiments

Introduction

Learning can often be thought of as a change in performance from experience. Sequence learning involves the individual becoming aware of the patterns or contingencies in the repeated pairing of events. The individual will begin to expect the next event to occur if they know that one event is likely to be followed by another. If there are a lot of repetitions of the pattern then the individual might perform quicker due to knowing the pattern without trying to actively memorise it.

One way to study sequence learning is the serial reaction time task. The individual will see a stimulus in one of the multiple positions and press the relevant key quickly and accurately. If there is a repeating pattern it can cause the individual to respond quicker due to knowing what will happen next. If the pattern is broken however, response times may increase. It has been shown that changes in response times due to sequences can be used as a measure of learning (Nissen & Bullemer, 1987). Later studies have shown that not always do individuals need to consciously know the pattern to improve performance (Willingham et al, 1989).

I created a 4 key press task using PsychoPy. I created two conditions, one with a 4 step repeating pattern and the other with a random pattern. The main aim of the experiment was to determine if the contingency in the repeating pattern would lead to quicker response times. I also wanted to look at how the response times changed throughout the trials.

Aim -To investigate if spatial key press contingencies would lead to quicker key press responses when there was a repeating pattern.

Hypothesis -I would expect there to be a lower mean response time in the sequential condition compared to the random condition.

Method

Data and Participants

I analysed 5 labelled datasets containing 400 trials. My own data was used to observe. The other datasets (Aarth, Dhruv, Vedanshi & Yash) were created as extra practice data sets. I was the participant recorded (20 year old undergraduate student). There was no demographic information available for the simulated participants.

Materials and Design

The experiment was designed using PsychoPy (2026.2.3). Two trials were created using the same routine (Peirce et al, 2019). Four white vertical lines were displayed on the screen indicating where the key press should take place. A triangle was displayed above one of the lines and the correct key pressed. The experiment had 4 horizontal positions (-150, -50, +50 & +150). The position of each line corresponded with the key press due to being set in the conditions within

the project. -150 was Z, -50 was X, +50 was C and +150 was V. The experiment was displayed in a full screen window of size 1024x768.
The trial began with a fixation cross for 1000ms. After 1000ms the lines were displayed and after 2000ms the triangle was displayed. The keyboard response component was used to record the key pressed, whether it was correct and the response time. The routine would end once a key press was made.

Procedure

The first loop was set to sequential and the second fullRandom. Both loops had 50 repetitions of the 4 row conditions file. This gave 200 trials for both the sequential and random conditions. The sequential condition had the pattern X-V-C-Z. This corresponded to the triangle appearing over the -50, +150, +50 & -150 positions. The random condition had the same 4 positions but in a random order. The task involved pressing the correct key quickly and accurately when the triangle appeared.

Data Analysis

Using key_resp.corr I could see which trials were correct (1) and incorrect (0). Using key_resp.RT I could see the response time in seconds. I calculated the mean response time for each condition. I only recorded the mean response time for correct trials. The response time was converted to milliseconds. I then calculated the difference between the mean response time of the random and sequential conditions with the sequential condition subtracted from the random. If the random condition had a higher value this meant it was slower. I also calculated the mean correct response time for each condition for 10 blocks of 20 consecutive trials. The seqRT and ranRT columns were helper columns so they were not counted as additional trials.

Results

There were 200 trials for each condition. In the sequential condition 192 trials were correct (96.0%) with a mean correct response time of 416.45ms. In the random condition 189 trials were correct (94.5%) with a mean correct response time of 413.27ms. The random condition had a -3.17ms difference. This was a very small difference and was the opposite to what was predicted.

Dataset	Correct, seq.	Seq. RT (ms)	Correct, rand.	Rand. RT (ms)	Rand − seq. (ms)
Hetansh	192/200	416.45	189/200	413.27	-3.17
Aarth *	191/200	502.57	177/200	516.55	+13.98
Dhruv *	187/200	499.45	187/200	532.79	+33.34

Vedanshi *	190/200	435.58	190/200	445.21	+9.63
Yash *	192/200	381.37	193/200	410.79	+29.42
Five-file average		447.08		463.72	+16.64

Table 1- Mean reaction time on correct trials. RT values are in milliseconds; positive differences indicate slower random responses.

For our five labelled files, the median of the individual means was 447.08ms for the sequential condition, and 463.72ms for the random condition. This gives us a total difference between the two means of +16.64ms. These extra numbers are to demonstrate how the calculation would be performed. They are not actual results from a sample of 5 people.

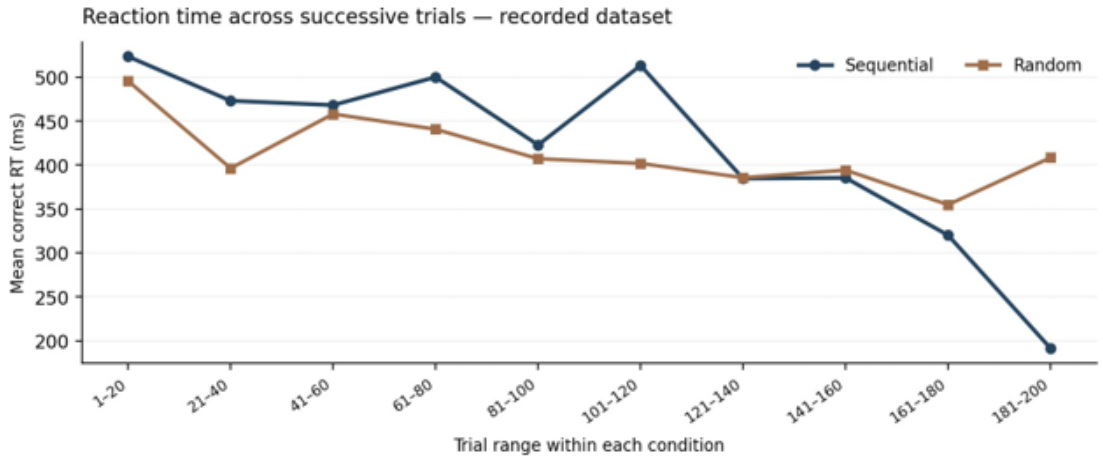


Figure 1 -Recorded participant's mean correct reaction time in successive groups of 20 trials, plotted separately for the sequential and random conditions.

The graph below highlights why just using the condition mean reaction times isn't enough. For the sequential condition there was a decrease in mean reaction time from 523.5ms (trials 1-20) to 191.3ms (trials 181-200). The reaction time for the initial 20 random trials was 495.6ms and decreased to 408.4ms for the last 20 trials. This can be seen as there was an increase of 304.3ms between the last block of trials for the sequential condition and the first block of trials for the random condition. This is just descriptive and doesn't help identify any sequence learning.

Discussion

As discussed earlier my hypothesis wasn't supported by the means of the condition. The random condition was faster than the sequential by approximately 3ms. This is a very small amount to make any meaningful claims. However throughout these trials there was a large improvement in my performance throughout the repeated condition and there was a noticeable increase in reaction times when moving to the random condition. This could be because I got used to the repeated sequence and needed to adapt to the new sequence. A more simple explanation could be that I just generally got quicker throughout the task with the keys. This could have been a factor.

Due to the nature of the sequential condition, I could predict where the next stimulus would appear as the sequence was repeated. Whereas for the random condition, whilst the positions were still familiar they weren't predictable. Previous research into serial reaction time tasks has highlighted that any changes in performance when the pattern is disrupted could indicate learning. It is however important to note that learning, awareness and simple motor practice aren't the same. (Nissen & Bullemer, 1987; Willingham et al, 1989). Howard et al (1992) looked at performance when switching from a repeated pattern to a random block. As I didn't measure awareness I can't confirm that learning was implicit.

A large issue with this experiment was that the conditions were always presented in the same order. Each file began with the sequential condition and then the random. By the time the random condition began the participant would have completed 200 key presses. The practice could have made them faster with their motor skills overall. However they may have got tired or been caught out by the sudden change in sequence. To help determine the cause of this issue counterbalancing could be used. About half the participants would have completed the sequential condition first and then the random condition. If the repeated pattern shows an advantage in reaction times for both groups it would be less likely to be due to it being the first or second condition.

There are also some issues with the analysis. Both conditions only had 4 possible positions to respond to. The sequence was only 4 steps long and was repeated 50 times. This creates a very limited task and only looks at one session. There was also no delayed retest meaning we can't determine if there was any retention/consolidation. Most importantly there was only one set of data collected from a participant (my file). Whilst the 4 practice sets help to show the descriptive analysis they don't increase the number of participants observed.

Conclusion

Overall in my recorded session the mean correct reaction time for the sequential condition was 416.45ms and for the random condition was 413.27ms. Although there was a small difference overall and my hypothesis wasn't supported there was a large change in reaction times throughout the 400 trials. This exercise shows how a repeated contingency can affect

performance and how you need to consider the progression throughout the trials and order of conditions to conclude that the difference in means is due to sequence learning.

References

- Howard, J. H., Mutter, S. A., & Howard, D. V. (1992). Serial pattern learning by event observation. Journal of Experimental Psychology: Learning, Memory, and Cognition, 18(5), 1029.
- Nissen, M. J., & Bullemer, P. (1987). Attentional requirements of learning: Evidence from performance measures. Cognitive psychology, 19(1), 1-32.
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., ... & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. Behavior research methods, 51(1), 195-203.
- Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., ... & Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. Behavior research methods, 51(1), 195-203.
- Willingham, D. B., Nissen, M. J., & Bullemer, P. (1989). On the development of procedural knowledge. Journal of experimental psychology: learning, memory, and cognition, 15(6), 1047.



Figure 2. Screenshot of the four-position display and triangular probe provided with the experiment.